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Control of prepotent responses by the superior medial frontal cortex

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ABSTRACT

The inhibitory control of prepotent action is vital for appropriate behaviour. An example of the importance of such control can be seen in the inhibition of aggressive behavior, deficits in which may have broader consequences for society. Many studies have related lesions or the under-development of the prefrontal cortex to inefficiency of inhibitory control. Here we used transcranial magnetic stimulation and a stop-signal task, which occasionally requires the inhibition of a prepotent motor response, to investigate the role of pre-supplementary motor area (Pre-SMA) in inhibitory control. While no effects were seen on the ability to generate responses, TMS delivered over the Pre-SMA disrupted the ability to respond to a stop signal. These results are the first to establish a casual link between Pre-SMA and inhibitory control in normal subjects. The understanding of the underlying mechanisms of inhibitory control may lead to clearer understanding of the neural basis of inappropriate behaviour.

Introduction

The control of voluntary action involves not only choosing from a range of possible actions but also the inhibition of responses as circumstances demand. The ability to inhibit prepotent responses is important to prevent execution of a behaviour in circumstances where to do so may be detrimental. Oft mentioned examples of such behaviour include the withholding of responses by batsmen in sports such as cricket and baseball, when there is a very brief period in which the choice must be made whether to make a motor response or not.

This behaviour can be investigated experimentally using stop-signal tasks. These involve the presentation of a target to which subjects have to respond unless a (relatively rare) stop signal is presented. Performance of such tasks is usually described in terms of a race between the go response and the stop response, with whichever reaches its threshold first governing the response made (Logan and Cowan, 1984; Logan et al., 1984; Boucher et al., 2007). Alteration in the time of onset of the stop signal allows for variation in the probability of responding in the trials with stop signals. The distribution of go reaction times and the probability of responding in the trials with stop signals can then be used to estimate the time required to inhibit the planned response, namely the stop-signal reaction time (SSRT).

The stop-signal task can be used to reliably estimate the response time of an internally generated act of control. This task therefore has been applied to investigate the underlying causes of some clinical syndromes related to impulsivity control. Deficits in performance on this task have been seen in attention-deficit hyperactivity disorder and conduct disorder (Schachar and Logan, 1990; Schachar et al., 1993, 1995; Armstrong and Munoz, 2003), Tourette's syndrome (Li et al., 2006a) and cocaine-dependent men (Li et al., 2008). This task has also been used to gauge the impulsivity of certain groups of subjects. For instance: Logan et al. (1997) found that highly impulsive subjects showed longer stop signal reaction times on this task. A similar line of study has shown that impulsive-violent offenders have longer SSRTs, requiring a longer time to inhibit their actions, compared to matched controls (Chen et al., 2008). These studies have demonstrated the wide utility of the task and its robust effects.

Recently, many studies have been focused on the neural correlates of the inhibitory processes recruited in the task. Evidence from both electrophysiological studies in non-human primates and human neuroimaging data have indicated that several frontal cortical regions are involved in the cognitive processes required for successful performance of the stop-signal task. These include frontal eye fields (FEF, Hanes and Schall, 1996), supplementary eye fields (SEF, Stuphorn et al., 2000; Stuphorn and Schall, 2006), anterior cingulated cortex (ACC, Ito et al., 2003; Chevrier et al., 2007), inferior frontal gyrus (IFG, Aron et al., 2003; Leung et al., 2007), and the pre-supplementary motor area (Pre-SMA, Li et al., 2006b). Several comprehensive papers have further illustrated the neural mechanisms of the inhibitory processes

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involved in the paradigm in regards of both awake-behaving monkeys' data (e.g. Schall et al., 2002; Schall and Boucher, 2007; Isoda and Hikosaka, 2007) and evidence from human neuroimaging studies (e.g. Aron et al., 2007a). With neurodisruption techniques, it is now established that both SEF (Stuphorn and Schall, 2006) and IFG (Chambers et al., 2006, 2007) are critical for inhibitory control. Stuphorn and Schall (2006) showed that such SEF stimulation increases the probability of countermanding a saccade, but there was no effect when no stop signal occurred. Chambers et al. (2006, 2007) first demonstrated that IFG is critical when manual responses are required in the stop-signal paradigm and showed that IFG is selectively necessary for inhibition and the SMA only critical for motor initiation.

In comparison with the findings of IFG and SEF involvement in response inhibition, whether the Pre-SMA has a critical functional role in this process is relatively less understood. The role of the Pre-SMA in action control has been investigated in several studies. For example, the Pre-SMA is involved when subjects must switch between making manual or saccade responses or when inhibiting such responses (Rushworth et al., 2007; Taylor et al., 2007). Imaging studies have implicated this area in the initiation of voluntary responses (Lau et al., 2004a,b). It is also involved in switching between tasks (Rushworth et al., 2002a,b) and between action sequences (Kennerley et al., 2004). This area is one of several (others of which include SMA and SEF) which have been implicated in self initiated or goal driven actions and it has been argued that voluntary selection of actions involves suppression of automatically triggered actions (Sumner et al., 2007). In recent years, several studies have also investigated the functional role of the Pre-SMA in response inhibition. Li et al. (2006b) used the duration of the SSRT in a stop-signal task to index the efficiency of inhibitory control and to investigate the functional role of the Pre-SMA in this control by use of fMRI. They compared groups of subjects with short and long SSRTs and found that higher activation in the Pre-SMA was associated with shorter SSRTs (i.e. more efficient performance). This was argued to be consistent with Pre-SMA having a role in mediating the motor inhibition for voluntary action (Nachev et al., 2005; Aron and Poldrack, 2006; Aron et al., 2007b). In addition, human patient studies revealed that damage to right superior frontal regions (including Pre-SMA and SMA) elevated subjects' SSRT in the stop-signal paradigm (Floden and Stuss, 2006, for a review see Mostofsky and Simmonds, 2008). Picton et al. (2007) also observed that patients with damage to the left superior regions of Brodmann area 6 (i.e. in the vicinity of the left Pre-SMA) showed increased numbers of incorrect responses to the nogo stimulus in a go–nogo task. The pattern of the results suggests that damage in the vicinity of the left Pre-SMA may impair inhibitory control in patients. However, it is not yet clear whether the Pre-SMA plays a critical role in inhibitory control in healthy human subjects. This is especially important since the findings of human patients studies may also involve an unknown degree of reorganization in the damaged brain which may occur rapidly. In particular, the Pre-SMA has direct connections to both the rIFG and the subthalamic nucleus, which are considered to be the main neural substrates for inhibitory control (Aron et al., 2007b). The findings from the human patients studies may reflect the changes of the functional weights across the neural nodes in the inhibitory network after the Pre-SMA lesion but not the actual function of the Pre-SMA. Even though the Pre-SMA has been associated with several cognitive functions related to the control of action, its critical role in response inhibition has not been tested directly with normal subjects. Understanding of this critical role of the Pre-SMA may shed some lights on the mechanisms underlying the operation of the inhibitory neural network.

While neuroimaging studies provide evidence of areas in the brain in which activity correlates with task performance, this approach alone cannot establish a causal link showing that an area is essential for task performance (Li et al., 2006b) and the human patients studies cannot avoid the potential confounds of neural plasticity or reorganization which may occur following the lesion. We therefore used temporally precise transcranial magnetic stimulation (TMS), which allows investigation of whether an area is required for performance, to evaluate the contribution of Pre-SMA to stop signal performance. Rushworth and colleagues (Rushworth et al., 2002a; Kennerley et al., 2004) have demonstrated that high frequency repetitive TMS (rTMS, 10 Hz, 0.5 s., i.e.: 5 TMS pulses for 400 ms) over the Pre-SMA can impair subjects' performance on the switch of motor sequences. In this study, we employed 10 Hz rTMS over the Pre-SMA but the duration of rTMS was 100 ms (i.e.: 2 pulses). This rTMS protocol has been widely used to investigate the functional roles of the primary visual cortex (V1), the frontal eye fields (FEF) and the posterior parietal cortex (PPC) in visual search tasks. Juan and Walsh (2003) used this rTMS protocol over V1 and found that rTMS impaired visual search function by decreasing subjects' d' score in two discrete time windows (for review see Juan et al., 2004; Chambers and Mattingley, 2005). A similar approach has been used to investigate the critical time windows of FEF involvement in visual search (O'Shea et al., 2004), to elucidate the different temporal involvements between FEF and PPC (Kalla et al., 2008) and to probe temporal dissociation between visual selection and saccade preparation in FEF (Juan et al., 2008). Both O'Shea et al. (2004) and Kalla et al. (2008) found that rTMS FEF and rTMS PPC decrease subjects' d' score and Juan et al. (2008) found that subjects' saccade latencies were prolonged by rTMS FEF in two separated time windows which represented the stages of visual selection and saccade preparation, respectively. Because Rushworth et al demonstrated that Pre-SMA rTMS interfered the function motor switching and the abovementioned findings of neurodisruptive effects of the rTMS protocol, we therefore hypothesized that the rTMS delivered over Pre-SMA would disrupt stop signal performance and such disruption would be manifested as modulation of the ability to respond to the stop signal (i.e. inhibitory control would be affected) rather than affecting action execution. The present study represents the first attempt to use rTMS to test the functional necessity of Pre-SMA in the inhibition of the prepotent responses with the stop-signal paradigm.

Aron et al. (2007a,b) suggested that the Pre-SMA, IFG and STN are parts of a neural network for the inhibitory control in stop-signal paradigm. Recently, Chambers and colleagues (2006, 2007) have used TMS over IFG and found the rate of noncancelled responses following a stop signal was increased by TMS (i.e.: inhibitory control was affected). Their findings have successfully established a causal role for IFG in the inhibitory control. The TMS protocol used in their studies was so called 1 Hz repetitive TMS stimulation. rTMS pulses were delivered for 10~15 min at a 1 Hz rate. The merit of this protocol is that it produces rTMS effects which last beyond the period of stimulation (for around 15 min) so subjects can be stimulated first then tested. However, the TMS effects of such a protocol are less event-related. In this study, because subjects received rTMS while they were performing the task, any rTMS effects observed here can reflect the “online” cognitive processes recruited by the task. The current study therefore not only probes the critical involvements of Pre-SMA in the inhibitory control of the stop-signal task but also the event-related rTMS effects over the stop-signal task.

Materials and methods

Participants

Nine volunteer college students (aged 21 to 35 years, mean 25.7, 7 male, 2 female, all right handed) took part in the experiment. All gave informed consent prior to participation. The experiment was approved by Institutional Review Board of the Veterans General Hospital, Taipei.

Apparatus

Testing took place in a sound attenuated room. Stimuli were presented on a 19-inch CRT screen using video resolution of

800×600 pixels and a vertical refresh rate of 100 Hz. The subjects sat 75 cm in front of the screen which was at eye level. The task was programmed using E-prime running on a Pentium IV PC which controlled the presentation of the stimuli as well as recording response information.

Procedure

Stop-signal task

In the stop-signal task, the stop signal delay (SSD) is the most critical independent variable and it is manipulated by adjusting the time between the onset of the go stimulus and that of the stop signal (Logan, 1994). The outcome of the race between the go process and the stop process is reflected by the inhibition function. This describes the probability of responding given a stop signal delay in accordance with the race model of Logan and Cowan (1984). The stop signal reaction time (SSRT) represents the latency of the stop process and it is the most important dependent variable in the task. The SSRT can be estimated from the observed distribution of RTs in no-stop signal trials in combination with the inhibition function (Logan, 1994). According to Band and colleagues' comprehensive review of the stop-signal paradigm, several methods are available for estimating the SSRT (Band et al., 2003; see also Logan, 1994). In the current study, SSRTs for each SSD were estimated using the integration method and one summary SSRT was calculated by averaging the three SSRTs acquired in three SSDs in our experiments (Logan, 1994; Band et al., 2003).

In the current study, each trial of the stop-signal task began with a central fixation dot which appeared for 500 ms. Following offset of this dot, a white target dot was presented to the left or right of the fixation at 9° of eccentricity on the horizontal meridian (see Fig. 1). On 75% of trials (go trials) subjects were required to make a key-press response on a response box with the left index finger when the dot was presented on the left, or with the right index finger when the dot was presented on the right. On 25% of the trials (stop trials), the central fixation dot reappeared and acted as an instruction to withhold responses to the peripheral target.

Baseline parameters

In order to reduce the number of trials in task and thereby the number of TMS pulses in the formal experiments, each subject's Mean RT for go (no-stop-signal) trials and critical SSD were acquired with three pre-TMS sessions (see Fig. 2). Every subject started with a session of the choice RT task (50 trials). Subjects were asked to respond to a target which appeared in the left or the right visual field with their corresponding index fingers. They were encouraged to make the responses correctly and as quickly as possible. The purpose of this session was to obtain each subject's mean go RT and standard deviation in the absence of stop signals. Each subject's mean go RTs plus two standard deviations was set as his/her time limit for go RT trials in the subsequent sessions. If the subject did not respond quicker than his/her time restriction in a go trial, the trial was counted as a non-responding error and a warning beep was delivered. It has been demonstrated that this procedure can effectively limit the strategy of slow responses to avoid errors, which subjects may use to reduce the rate of non-cancelled errors (Chen et al., 2008). Floden and Stuss (2006) also used a similar procedure to evaluate baseline go RT and to restrain strategic slowing. Moreover, Garavan et al's study (2002) suggested that the pre-SMA was more involved in inhibitory control when the time pressure was present in the go-nogo task. We therefore used this procedure to reduce strategic slowing and to increase the involvement of the pre-SMA in the task.

A practice session which was composed 24 go trials and 8 stop trials followed the choice go RT session. The SSD was fixed at 170 ms in the stop trials in this session, but the experimental sequence of the trials in this session was otherwise identical to the abovementioned sequence and the subsequent formal TMS sessions. After the subjects performed the go time restricted session and the practice session, they were required to carry out a critical SSD session (see Fig. 2). The

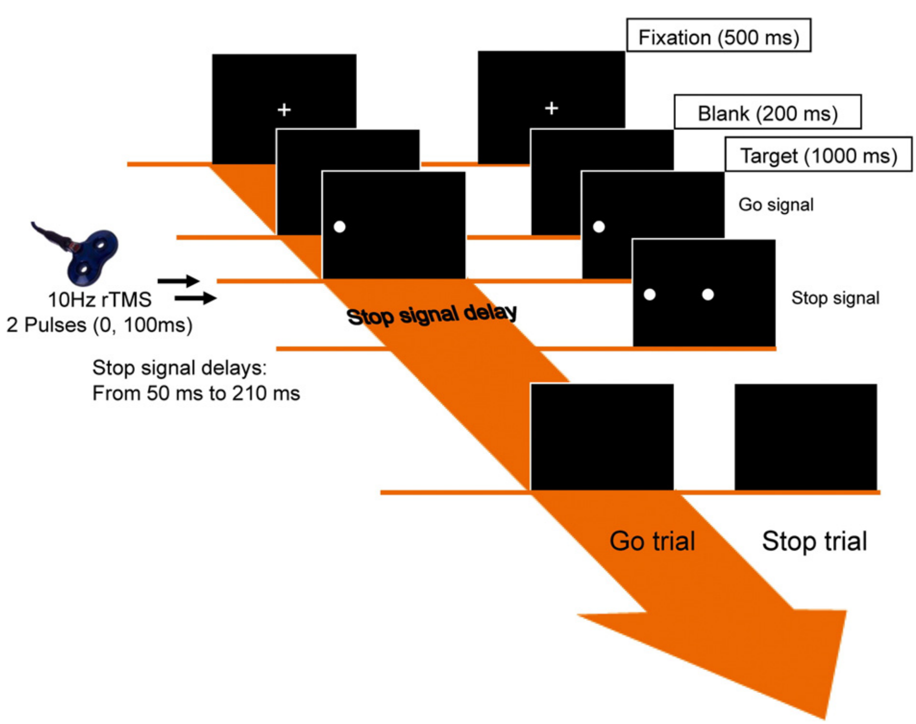

Fig. 1. Task procedure. The stop-signal task consisted of go and stop trials. All trials began central fixation. Following offset of the central fixation, a white peripheral dot was presented to the left or right of the fixation. Subjects were required to make a keypress response on a response box with the left index finger when the dot was presented on the left or with the right index finger when the dot was presented on the right. On 25% of the trials (stop trials), the central fixation dot reappeared as an instruction to withhold responses to the peripheral target.

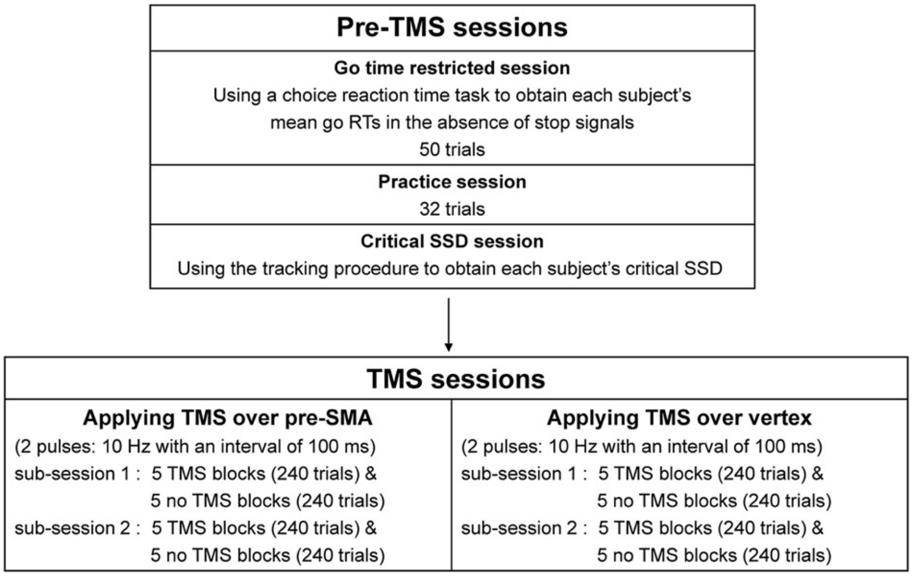

Fig. 2. The procedures of the experimental sessions. Three pre-TMS sessions were carried out to establish each subject's baseline RT and to acquire his/her critical SSD. Subjects then received two TMS sessions. The order of the two TMS sessions were counterbalanced across subjects.

purpose of this session was to estimate every subject's SSD at which their non-cancelled rate would be around 50%. This session also helps to decrease the number of trials in the formal TMS sessions since the critical SSD does not then have to be obtained *during* the TMS trials and there is then a better chance to observe the effects of TMS (see also Chambers et al., 2007). The tracking procedure was used for acquiring the critical SSD. According to the results of our pilot experiments, the initial SSD was set at 170 ms. The SSD of each subject was adjusted until the subject's accuracy on stop-trials reached 50%. The program monitored subjects' performance block by block. If the subject's non-cancelled rate was lower than 37.5%, the SSD was increased by 40 ms. Conversely, if the non-cancelled rate was higher than 62.5%, the SSD was decreased by 40 ms. A critical SSD could be computed that represented the time delay required for the subject to succeed in withholding a response in the stop trials half of the time. Subject's critical SSD was determined when their non-cancelled rate was within 37.5%~62.5% for two consecutive blocks. It usually took subjects less than 500 trials to obtain their critical SSD.

TMS blocks

The main body of the experiment consisted of three conditions; two of these involved TMS (over the left Pre-SMA and the vertex) and one with no TMS. Three SSDs were presented to each subject based on their individual critical SSDs; critical SSD, 40 ms less than critical SSD, and 40 ms more than critical SSD. For example, if a subject's critical SSD was 210 ms (acquired in the Critical SSD session), the other two conditions were 170 ms and 250 ms. During blocks with TMS, two pulses with an inter-pulse interval of 100 ms were delivered over the relevant stimulation site concurrent with the onset of the go signal.

Each experimental block included 48 trials and lasted approximately 4 min; the occurrence and order of the three stop signal presentation conditions was randomized within each block. For each TMS site, subjects received 20 blocks of trials, of which 10 blocks were with TMS and the other 10 blocks were without TMS, serving as control trials. An ABBA design was used to control for sequence effects. Each TMS session was divided into two sub-sessions. Subjects received 5 TMS and 5 control blocks totalling 10 blocks in the first sub-session. After a break of 10 min, subjects received the second sub-session of 10 blocks. Subjects only received one TMS session in a day and received the second TMS session a week after the first. Subjects were randomly assigned to receive either the Pre-SMA TMS or the vertex session as their first TMS session (see Fig. 2).

TMS parameters and site localization

A Magstim Super-Rapid Stimulator was used to deliver TMS at 60% of maximum machine output (approx. 1.2 Tesla, duration of one pulse: less than 1 ms) over Pre-SMA and the vertex. A fixed stimulation level was used because it has proven successful and replicable in many studies and over a wide range of tasks (e.g.: Ashbridge et al., 1997; Rushworth et al., 2002a,b; Muggleton et al., 2003; Hung et al., 2005; Ellison and Cowey, 2007, Juan et al., 2008) and because motor cortex excitability does not provide a good guide to TMS thresholds in other cortical areas (Stewart et al., 2001). Stimulation was delivered via a 70 mm figure of eight coil held clamped in position with the handle parallel to the sagittal midline (with the direction of the current over the stimulation site travelling in the same lateral to medial direction).

The stimulation site for the Pre-SMA was localized in each subject using a magnetic resonance image (MRI)-guided frameless stereotaxy system (Brainsight, Rogue Research, Montreal, Canada). Briefly, identification of the Pre-SMA site on the structural scans was achieved by the following procedure. Individual MRIs were normalized against a standard template using the FSL software package (FMRIB, Oxford). This produced a matrix describing the transformation applied to the structural scan to result in the normalised brain. This was then reverse-applied to the coordinates for Pre-SMA (−4, 32, 51, Li et al., 2006b see Fig. 3) to obtain the location of the site in the original structural scan for each subject. The location was then marked on the MRI scan in the Brainsight system.

After the Pre-SMA location had been identified in a subject's structural MRI scan, a Polaris infra-red tracking system (Northern Digital, Waterloo, Canada) was then used to co-register the positions of anatomical landmarks on each subject's head which were also visible on each MRI scan (bridge of nose, nose tip, left and right intra-trageal notches). Another infra-red tracker was placed over the TMS coil and was utilized to identify the scalp point over the identified Pre-SMA site. For vertex stimulation the coil was held anterior to the handle which was oriented parallel to the sagittal midline. This site was localized by marking the point midway between the intertragal notches and midway between the inion and nasion. The scalp positions of the Pre-SMA and the vertex were marked on a cloth swimming cap which was worn throughout the test sessions.

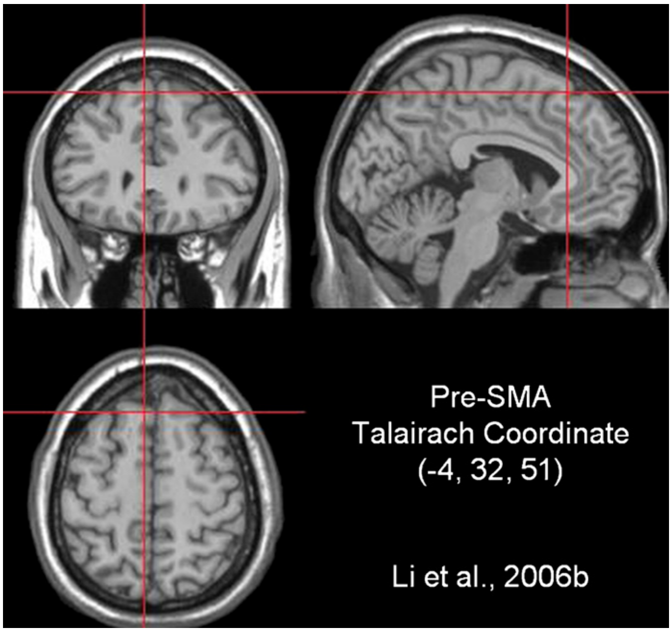

Fig. 3. Site localization. The Pre-SMA magnetic stimulation sites were localized using the Brainsight TMS-MRI co-registration system (Rogue Research, Montreal, Canada). The Tailarach coordinates of −4, 32, 51 reported by Li et al. (2006b) were used and lay in as the superior medial frontal cortex. The vertex was defined as a point midway between the inion and the nasion and equidistant from the left and right intertrachial notches.

Analysis

The go RTs were filtered by removing non-response trials, trials with responses to the wrong target and trials with latencies below 200 ms. In addition, trials with latencies more than 2 standard deviations away from each subject's mean Go RT in the baseline GO session were also excluded from further analysis. When TMS was applied over pre-SMA, the error rates for non-response trials, trials with responses to the wrong target, the proportions of trials with latencies below 200 ms and GO RT above the mean plus 2 SD were 0.002, 0.003, 0.017 and 0.046, respectively. The mean GO RT plus 2 SD across subjects was 421 ms (SD of 82 ms).

The same procedures were also applied to the Vertex TMS trials. When TMS was applied over vertex, the error rates for non-response trials, trials with responses to the wrong target, trials with latencies below 200 ms and GO RT above the mean plus 2 SD were 0.003, 0.002, 0.008 and 0.03, respectively. The mean of GO RT plus 2 SD across subjects was 425 ms (SD was 53 ms). These trials were excluded from further analysis.

For the no TMS condition, the same exclusion criteria were again applied and the error rates were 0.004, 0.001, 0.014 and 0.048, respectively. The mean GO RT plus 2 SD across subjects was 413 ms (SD was 35 ms).

Individual mean reaction times for the correct trials were analyzed after removal of the abovementioned trials. The estimation of the internal response time to the stop signal (SSRT) was calculated using the distribution of go signal reaction times and the probability of responding given a stop signal delay (the inhibition function) in accordance with the race model of Logan and Cowan (1984). The inhibition function described the probability of responding (noncancelled rate) as a function of SSD. The latency of the stop process can be estimated from the RTs on no-stop-signal trials (the observed go RT distribution) in combination with the inhibition function. In this study, SSRTs for each stop signal delay were estimated using the integration method (Hanes et al., 1998; Hanes and Carpenter, 1999) then averaged to obtain a summary SSRT (i.e.: $SSRT_{average}$ in Band et al's terminology (2003)). We followed the method introduced in Logan (1994; see also Logan and Cowan, 1984; Band et al., 2003) to calculate the SSRT in each SSD. Briefly, if the non-cancelled rate $=x$, at a given SSD, the stop processes must have finished at point x of the observed go RT distribution. The value of the x point minus SSD yields the SSRT. For example, if SSD = 130 ms, non-cancelled rate = 0.4, and the 40th percentile RT of the observed go RT distribution = 330 ms, the

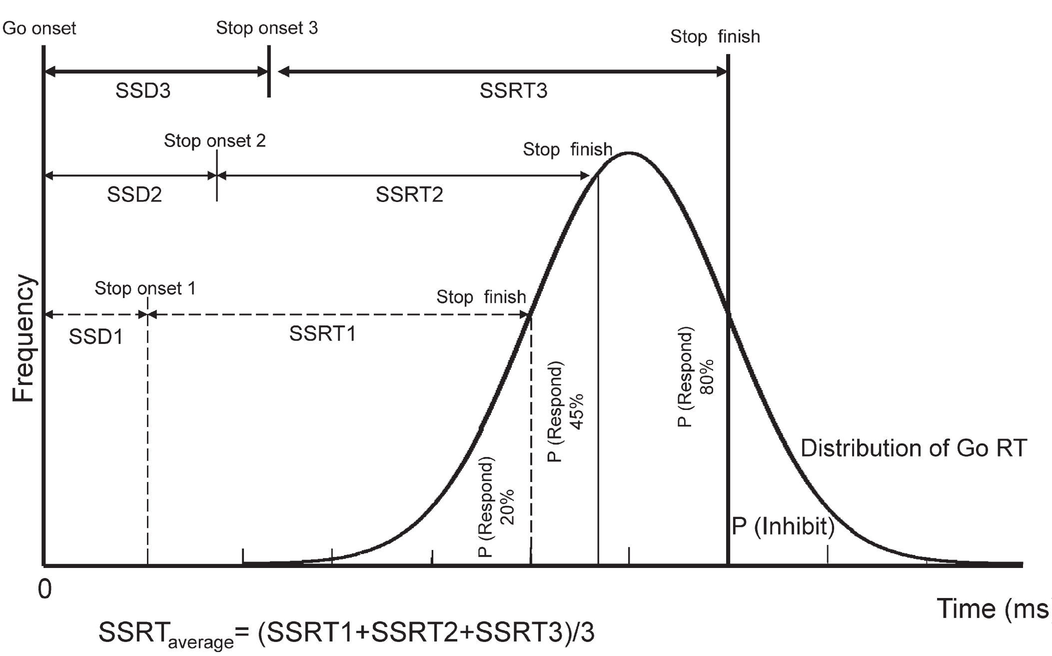

Fig. 4. Stop signal reaction time calculation. The figure shows the relationship between stop signal delay, the stop signal reaction time, and the distribution of go reaction times. The distribution of go reaction times is integrated from the time of go signal presentation. For each stop-signal delay, a probability of responding is obtained. If the stop-signal delay of 50 ms resulted in an error rate = 0.20, this means that the end of the stop process should be at a point equal to 20% of the go RT distribution. If the point of 20% of the go RT distribution was 252 ms, so the observed SSRT would be 252 − 50 = 202 ms. The rest of the SSRTs were calculated with the same procedure. A summary SSRT was acquired by averaging the observed three SSRTs that corresponded to $0.15 < p\,(\text{respond}) < 0.85$ (Band et al., 2003).

observed SSRT will be 330–130=200 ms for this SSD. Although this method assumes that the SSRTs are constant across trials, violation of this assumption does not significantly alter the outcome of the analysis (Logan and Cowan, 1984; Hanes et al., 1998). Notwithstanding this, Band et al. (2003) found observed SSRTs changed with stop-signal delays. Consequently, we calculated the average SSRT for stop signal delays where the proportion of trials in which subjects failed to inhibit responding lay between 0.15 and 0.85 (Band et al., 2003). This range of the inhibition function was selected because the slope of this function can be approximated as a straight line. Outside this range, ceiling and floor effects may result in a shallower slope of the inhibition function (Band et al., 2003). Fig. 4 shows an example how a summary SSRT was acquired in one subject.

Results

Repeated measures analysis of variance was carried out for go RT (for correct and noncancelled trials), SSRT and accuracy with factors of TMS site (Pre-SMA, vertex and no TMS), and response hand. As there was no significant effect of response hand ($F_{(1,8)}$=1.559, P=0.247) (see Supplementary Information A for a comparison of the performance of the two hands). The data were collapsed for this factor and the analysis repeated with it omitted. The descriptive data are summarized in Table 1.

Go RTs (correct responses)

Fig. 5a shows the mean go RTs. There were no significant effects of TMS condition on go RTs (pre SMA 285.7±24.9, vertex 284.2±21.6, no TMS 285.5±29.8 ms, $F_{(2,16)}$=0.113, P=0.894).

Go RTs (noncancelled responses)

Fig. 5b shows the mean go reaction times when responses were not inhibited appropriately. Again there was no significant effect of TMS on go RTs for these responses (pre SMA 269.5±20.7, vertex 269.6.2±18.4, no TMS 268.9±21.3 ms, $F_{(2,16)}$=0.029, P=0.971).

Mean error rates

Fig. 5c shows the noncancelled rates. Significant differences were observed among the noncancelled rates for the three TMS conditions ($F_{(2,16)}$=4.071, P=0.037). The mean noncancelled rate when TMS was applied to Pre-SMA (0.65±0.14) was significantly higher than when applied to vertex (0.56±0.12, $t_{(8)}$=2.396, P=0.043) and also higher than in the no TMS condition (0.57±0.11, $t_{(8)}$=2.565, P=0.033).

Table 1

	Go RT (ms)	% Go	% Noncancelled	SSRT (ms)
Pre-SMA				
Total	285.7±24.9	93.8±4.8	64.8±13.4	210.5±15.7
Sub-session 1	289.2±28.5	95.3±3.4	68.9±12.2	219.5±14.6
Sub-session 2	284.8±26.2	92.4±7.8	60.4±15.5	202.8±15.2
Vertex				
Total	284.2±21.6	95.7±1.3	56.3±12.1	193.7±15.1
Sub-session 1	284.8±20.2	95.5±3.6	54.6±12.5	191.3±13
Sub-session 2	285.6±24.7	96.1±2.7	57.8±12.4	200.1±15.8
No TMS				
Total	285.5±29.8	93.4±2.4	55.2±10.7	196.9±11.2
Sub-session 1	285.9±31.5	93.6±3.1	57.5±10.9	198.1±6.9
Sub-session 2	286.6±29.0	93.1±2.9	56.1±11.6	195.5±8.7

% Go=percentage of successful go trials.
% Noncancelled=percentage of noncancelled trials.
SSRT=stop-signal reaction time.
All numbers are mean±standard deviation.

Inhibition function

Fig. 5d shows the inhibition function for the different TMS conditions. The noncancelled rate was significantly increased with the increment of SSDs. ($F_{(2,16)}$=131.8, P=0.000). The main effect of TMS condition was significant ($F_{(2,16)}$=6.367, P=0.009). The post hoc results showed that the Pre-SMA TMS significantly increased the noncancelled rate in comparison to Vertex TMS ($t_{(8)}$=2.396, P=0.044) and NO TMS ($t_{(8)}$=2.565, P=0.033) conditions. The interaction between the SSD factor and the TMS conditions factor was not significant ($F_{(4,32)}$=2.11, P=0.1).

Stop signal reaction times

Fig. 5e shows the mean SSRTs for the different TMS conditions. There were significant differences across the different conditions ($F_{.(2,16)}$=9.866, P=0.002). The SSRT for applying TMS to Pre-SMA (210.5±15.7 ms) was significantly longer than applying TMS to vertex (193.7±15.1, $t_{(8)}$=3.505, P=0.008) and no TMS condition (196.9±11.2, $t_{(8)}$=2.917, P=0.019).

In the study of the role of IFG in inhibitory control by Chambers et al. (2006) they found a significant effect of session on the modulation of performance by TMS. For this reason we analysed the data described above with an additional factor of sub-session order.

The go reaction times of first-TMS and second-TMS sub-sessions separately

Fig. 6a shows the mean go reaction times of 1st TMS sub-session and 2nd TMS sub-session for three stimulation site conditions. Analysis of go RTs used ANOVA with sub-sessions and stimulation sites as factors. There was no interaction between the go RTs of the two sub-sessions across the different stimulation site conditions ($F_{(2,16)}$=0.759, P=0.484). There was no main effect for stimulation sites ($F_{(1,8)}$=0.175, P=0.841). There was also no main effect for sub-sessions ($F_{(1,8)}$=0.357, P=0.567).

The mean reaction times of noncancelled trials of 1st TMS and 2nd TMS sub-sessions separately

There was no interaction between the mean reaction times of noncancelled trials of the two sub-sessions across the different stimulation site conditions ($F_{(2,16)}$=0.428, P=0.659). There was no main effect for stimulation sites ($F_{(2,16)}$=0.423, P=0.662). There was also no main effect for sub-sessions ($F_{(1,8)}$=0.428, P=0.659) (Fig. 6b).

The mean noncancelled rates of first-TMS and second-TMS sub-sessions separately

A significant interaction was observed between the mean noncancelled rates of the two sub-sessions across different stimulation site conditions ($F_{(2,16)}$=11.022, P=0.001) (Fig. 6c).

In the 1st TMS sub-session, there was a simple main effect for stimulation sites ($F_{(2,16)}$=9.025, P=0.002). The mean noncancelled rate for applying TMS to Pre-SMA (0.69±0.12) was significantly higher than applying TMS to vertex (0.55±0.13, $t_{(8)}$=3.924, p=0.004) and no TMS condition (0.58±0.11, $t_{(8)}$=3.892, p=0.005). The noncancelled rate of 1st TMS sub-session (0.69±0.12) was significantly higher than that of 2nd TMS sub-session (0.6±0.15) while TMS was applied to Pre-SMA ($F_{(1,8)}$=14.37, P=0.005). There was no significant difference between the 1st TMS and 2nd TMS sub-sessions while TMS was applied to vertex ($F_{(1,8)}$=1.933, P=0.202). There was also no significant difference between the 1st and 2nd sub-sessions in the no TMS condition ($F_{(1,8)}$=0.452, P=0.52).

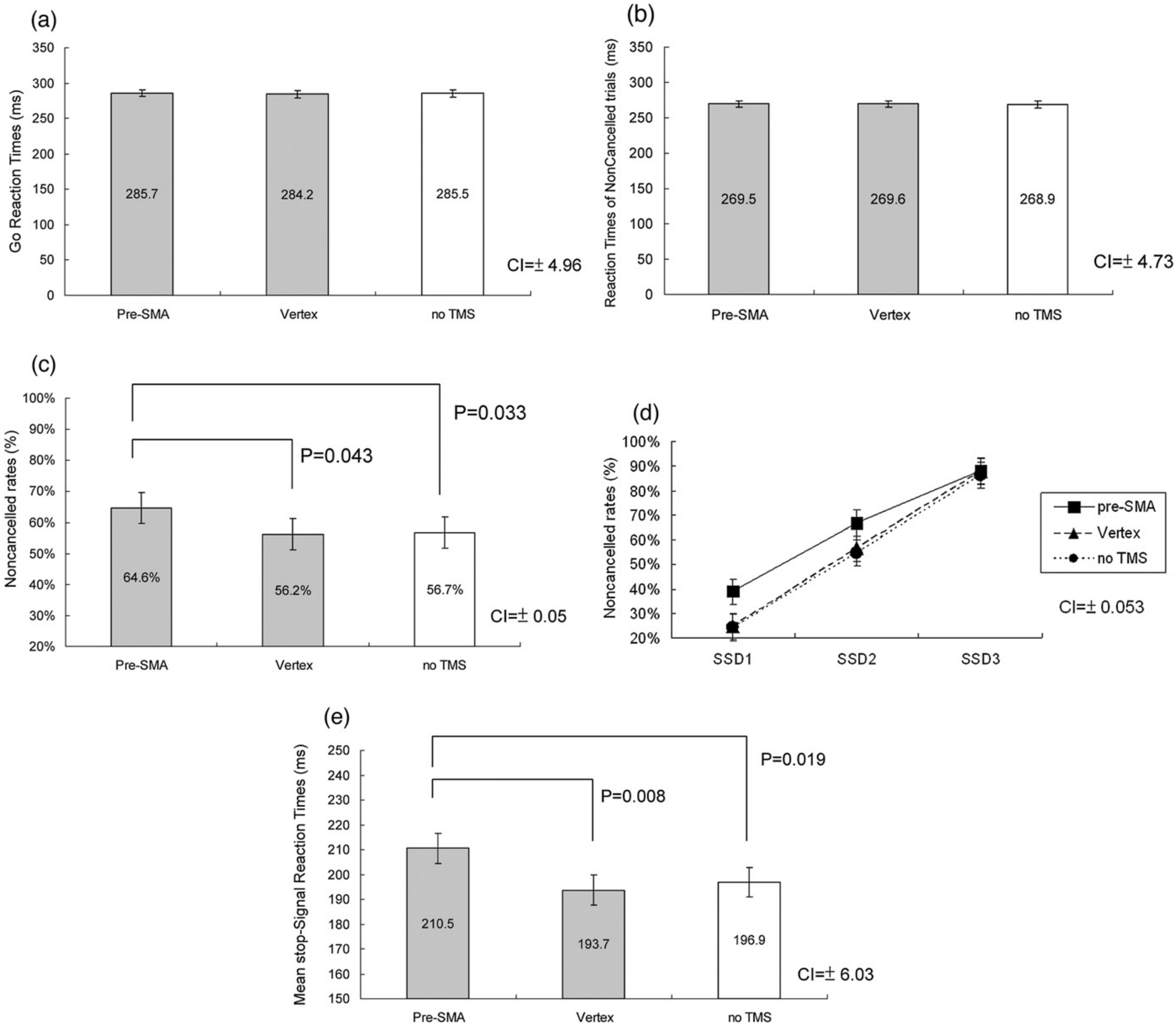

Fig. 5. Panel (a) to (e) show the mean go reaction times, mean reaction times of noncancelled trials, mean noncancelled rates, inhibition function of the three SSDs and stop-signal reaction times across three different conditions (TMS, vertex and no TMS). Error bars show a 95% confidence interval (CI). CI was computed on a within-subjects basis in accordance with Loftus and Masson (1994).

Stop signal reaction times of first-TMS and second-TMS sub-sessions separately

In the case of the SSRT variable, a significant interaction effect was seen between the SSRTs of two sub-sessions across three stimulation sites ($F_{(2,16)}=32.719$, $P=0.000$) (Fig. 6d). In 1st TMS sub-session, there was a simple main effect for stimulation sites ($F_{(2,16)}=30.274$, $P=0.000$). The mean SSRT for applying TMS to Pre-SMA (219.5±14.6 ms) was significantly longer than applying TMS to vertex (191.3±13 ms, $t_{(8)}=6.189$, $P=0.000$) and no TMS condition (198.1±6.9 ms, $t_{(8)}=5.563$, $P=0.001$). In 2nd TMS sub-sessions, there was no main effect for stimulation sites ($F_{(1,8)}=2.065$, $P=0.159$).

The SSRT of 1st TMS sub-sessions (219.5±14.6) was significantly longer than that of 2nd TMS sub-sessions (202.8±15.2) while TMS was applied to Pre-SMA ($F_{(1,8)}=11.928$, $P=0.009$). The SSRT of 1st TMS sub-sessions (191.3±13) was significantly shorter than that of 2nd TMS sub-sessions (200.1±15.8) while TMS was applied to vertex ($F_{(1,8)}=6.496$, $P=0.034$). There was also no significant difference between the pre- and post-sub-sessions in the no TMS condition ($F_{(1,8)}=2.282$, $P=0.169$).

In summary, our results indicated that TMS applied over the Pre-SMA significantly affected subjects' inhibitory control. The Pre-SMA TMS effects were reflected in the increase of noncancelled rates (Fig. 5c), the alteration of the inhibition function (Fig. 5d) and the increase of SSRTs (Fig. 5e). The Pre-SMA TMS effects seem more substantial in sub-session I (Figs. 6c, d).

Discussion

The ability to appropriately inhibit prepotent responses is necessary to prevent certain behaviours in circumstances where to do so may be associated with adverse consequences. Several areas in the prefrontal cortex have been associated with the mechanisms underlying such inhibitory control, with a network including left and right IFG, right MFG, ACC, Pre-SMA, FEF and right inferior parietal lobule all implicated by a range of studies (Aron et al., 2003; Chevrier et al., 2007; Leung et al., 2007; Hanes and Schall, 1996; Ito et al., 2003; Li et al., 2006b; Stuphorn et al., 2000; Stuphorn and Schall, 2006). It has previously been shown, using TMS, that the IFG plays a causal role in inhibitory control (Chambers et al., 2006, 2007). We investigated the role of another of the

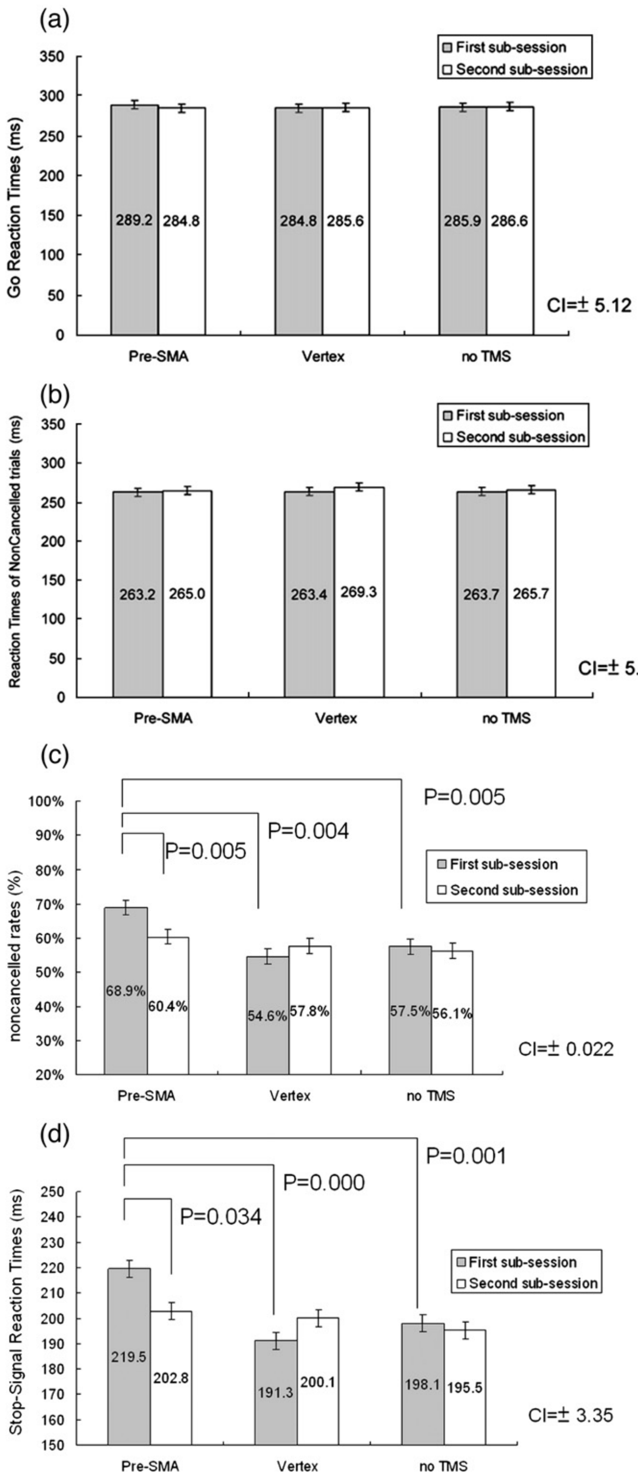

Fig. 6. Panel (a to d) show the mean go reaction times, mean reaction times of noncancelled trials, mean noncancelled rates and stop-signal reaction times of first-TMS and second-TMS blocks separately for three different conditions (TMS, vertex and no TMS). Error bars show a 95% confidence interval (CI).

implicated areas, the Pre-SMA, in inhibitory control. We tested the hypothesis that this area performs functions related to inhibition of responses, by use of a stop-signal task which allows measurement of both impulsivity and inhibition (Logan et al., 1997).

TMS delivered over left Pre-SMA resulted in effects consistent with this hypothesis, producing both elevated SSRTs and increased error rates compared to control stimulation. Both of these measures are indices of inhibitory control, and so the results are related to a pattern of disruption consistent with the findings of Li et al. (2006b). This study found that activation of the Pre-SMA was correlated with shorter SSRTs, which suggested that greater activity in Pre-SMA is indicative of more effective inhibitory control. Our data showed that TMS delivered over the left Pre-SMA resulted in deteriorated inhibitory control for both hands. This pattern of results may seem surprising on initial consideration. However, Isoda and Hikosaka (2007) recently applied microstimulation to monkeys' Pre-SMA and found that the Pre-SMA played a critical role in suppressing an unwanted action and facilitating a desired one. Intriguingly, they also found that a certain proportion of the recorded neurons responded to ipsi-lateral saccadic responses or even responded to bilateral saccadic responses. Similar findings were reported in Stuphorn and Schall's SEF microstimulation study (2006). Furthermore, subjects were asked to respond within a limited time period in the current study, and this may increase the involvement of Pre-SMA (Garavan et al., 2002) with Pre-SMA TMS therefore affecting motor selection and inhibition in both hands. It is also worth mentioning that the effect here may be a consequence of the medial location of the coil such that it might also activate the right Pre-SMA to some extent then cause the left hand effects, although this is considered unlikely.

The role of Pre-SMA in behavioural performance has been posited to be one of error monitoring (e.g. Klein et al., 2007). The findings here are inconsistent with this. The most obvious reason for conflict between our data and an error monitoring account is that TMS delivery occurs prior to a response. As such, the data are incompatible with TMS modulation of activity related to the accuracy of performance. TMS has high temporal specificity, indeed this is one of it's oft mentioned advantages, and consequently delivery during the trial is most likely to have effects, if any, on performing the trial effectively, rather than on the much later, in psychophysical terms, process of monitoring the outcome of the trial.

Even if TMS at this time point were to affect error monitoring, the pattern of effects seen are not what would be predicted. The disruption of feedback related to responses would be more inclined to lead to less conservative responding due to the reduced influence of making an error. This would be manifested in the results as a reduced go RT, in addition to increased error rates, and failure to inhibit responses. The data obtained here are consistent with the findings of Taylor et al. (2007). They found that Pre-SMA plays a direct and casual role in conflict resolution of an action selection task using a combination of TMS and electroencephalographic recording. Aron and colleagues (2007b) conducted a fMRI study with a conditional stop-signal paradigm and suggested similar functional role of the Pre-SMA in conflict resolution. It has also been suggested that the Pre-SMA is in charge of both response inhibition and response selection (e.g. Mostofsky and Simmonds, 2008). It would also be crucial to test whether the Pre-SMA becomes more important for response execution under conditions of increased response conflict.

Furthermore, our finding confirms the critical role of Pre-SMA within the neural network of the Pre-SMA, IFG and STN, which are fundamentally involved in inhibitory control in the stop-signal paradigm (Aron et al., 2007b). We found that the effects of TMS over Pre-SMA also were dependant upon familiarity with the task. Modulation of performance was only seen when the task was novel and was unaffected on the second testing session. This interaction between learning and TMS has been seen for other tasks, notably for conjunction visual search performance and TMS delivered over parietal cortex. Walsh et al. (1998) showed that TMS related disruption of conjunction search no longer occurred when subjects were well practiced at the task following learning over a period of days. Data obtained here might reflect Pre-SMA being more important in response inhibition during the formation of new (inhibitory) associations. Even if this is the case, it is noteworthy that there was no indication of a learning effect across sessions. Another possible cause of the session effects observed in the current study is that the functional weight of the Pre-SMA in inhibitory control may change across time. In other words, other neural nodes in the inhibitory network such as IFG and STN may be more important once the inhibitory associations are more familiar to the subject. Future work might fruitfully test these ideas further (see also Chambers et al., 2006). In conclusion, the Pre-SMA has been implicated in inhibitory control by a number of studies, both in monkeys and in humans. TMS delivered over this area showed, using a stop-signal task, that this area is necessary for effective inhibitory control.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.neuroimage.2008.09.005.

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